Tomas Petricek

http://tomasp.net | tomas@tomasp.net | 0000-0002-7242-2208

Academic education

- **Habilitation, Faculty of Mathematics and Physics, Charles University**, 2026
  Thesis Simple Programming Tools for Data Exploration uses programming language theory research methods to develop novel data science tools. The thesis summarizes research on F# type providers, data journalism and programming tools done in Microsoft Research, The Alan Turing Institute and at the University of Kent.
- **PhD, Computer Laboratory, University of Cambridge**, 2011 – 2016
  Thesis Context-aware Programming Languages develops coeffects, a theory for tracking information about environment in which programs are executed. I also devised a novel way of presenting the results in the form of widely read interactive essay (tomasp.net/coeffects). Supervised by Prof. Alan Mycroft.
- **BSc and MSc, Computer Science, Charles University, Prague**, 2004 – 2010
  MSc completed with distinction. Final thesis Reactive Programming with Events, supervised by Dr Don Syme, developed abstractions for reactive and concurrent programming that were presented in workshop papers.

Professional experience

- **Associate Professor, Charles University, Prague**, 2026 – present
- **Assistant Professor, Charles University, Prague**, 2022 – 2026
- **Lecturer, University of Kent**, 2018 – 2022
- **Collaborating Fellow and Visiting Researcher, The Alan Turing Institute**, 2016 – 2020
- **Post-doctoral Researcher and Contractor, Microsoft Research Cambridge**, 2014 – 2016

Research Funding

- **PRIMUS Research Programme, Charles University**, Award value €610,000 (2024 – 2028)
- **Fellowship, ACM History Committee**, Award value $4000 (2018)
- **Dstl & GCHQ grants, The Alan Turing Institute**, Award value £420,000 (2017 – 2020)
- **Innovation Fund, Google Digital News Initiative**, Award value €50,000 (2016)

Professional Membership

- **IFIP TC2 Working Group 2.16 Language Design** (2024 – present). Member of an invitation-based group of 40 leading programming language experts that meets twice a year for an intensive week-long meeting.

Awards

- **Reviewers' Choice Award**, ‹Programming› 2023. Awarded for a paper on programming systems.
- **Reviewers' Choice Award**, ‹Programming› 2020. Awarded for a paper on data exploration tools.
- **ACM SIGPLAN Research Highlight**. My first-author paper was chosen as an ACM SIGPLAN Research Highlight, making it one of the three best programming language papers of the past year.
- **Editors' Choice Award**, ‹Programming› 2018. Awarded for a paper on the history of monads.
- **Reviewers' Choice Award**, ‹Programming› 2017. Awarded for a paper on the history of errors.
- **Distinguished Paper Award**, PLDI 2016. Awarded for a paper on types for semi-structured data

Keynotes and Invited Talks

- **Undone Science in Computer Science**, Luxembourg (March 2026). Keynote: Undone Ideas on Programming
- **PLISS Summer School**, Bertinoro (June 2025). Invited Talk: Programming systems deserve a theory too!
- **Workshop of The Psychology of Programming Interest Group (PPIG)**, Belgrade, Serbia (September 2025). Keynote: Critical Architecture / Software Theory
- **Data Science in F#**, Berlin (September 2023). Keynote: Designing composable visualizations
- **ASL/APA Meeting**, San Francisco (April 2023). Invited Talk: Cultures of programming
- **Huawei Global Technology Summit**, Edinburgh (July 2022). Invited Talk: Rethinking data exploration tools
- **CodeMesh**, Virtual (November 2020). Keynote: Cultures of programming
- **ScalaDays**, Berlin (May 2018). Keynote: Functionalist programming language design
- **Data Science Summit**, Cambridge (June 2017). Keynote: The Gamma: Democratizing data science
- **ManLang Conference**, Prague (Sept 2017). Keynote: Language Challenges of Targeting Multiple Runtimes

Recent publications

For historical reasons, highly-selective conference publications are preferred over journal papers in computer science. I am the author of 20 papers in highly-selective venues (including POPL, PLDI, ICFP, ICALP, ECOOP) and 26 other peer-reviewed papers. I have an h-index of 16 (7 in WoS) and my work has 1080+ citations (233 in WoS).

- **Tomas Petricek**. Cultures of Programming: The Development of Programming Concepts and Methodologies. 420 pages, Cambridge Unviersity Press, ISBN 9781009492379, 2026
- **Tomas Petricek**, Jonathan Edwards. Denicek: Computational Substrate for Document-Oriented End-User Programming. Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology (UIST '25), no. 32, pp. 1-19, 10.1145/3746059.3747646, 2025
- **Tomas Petricek**, Joel Jakubovic. On the Limits of Making Programming Easy. Languages, Compilers, Analysis-From Beautiful Theory to Useful Practice: Essays Dedicated to Alan Mycroft on the Occasion of His Retirement, LNCS, vol. 15500, Springer, 10.1007/978-3-032-08187-2_11, 2025
- Peter Taylor-Gooby, **Tomas Petricek**, Jack Cunliffe. Covid-19, Charitable Giving and Collectivism: A Data-Harvesting Approach. Journal of Social Policy, vol. 52, issue 3, pp. 473-494, Cambridge University Press (IF: 1.9), 10.1017/S0047279421000714, 2023
- Joel Jakubovic, Jonathan Edwards and **Tomas Petricek**. Technical Dimensions of Programming Systems. The Art, Science, and Engineering of Programming, vol. 7, issue 3, no. 13, 10.22152/programming-journal.org/2023/7/13, 2023
- **Tomas Petricek**, Gerrit J. J. van Den Burg, Alfredo Nazábal, Taha Ceritli, Ernesto Jiménez-Ruiz and Christopher K. I. Williams. AI Assistants: A Framework for Semi-Automated Data Wrangling. IEEE Transactions on Knowledge and Data Engineering, vol. 35, issue 9, pp. 9295-9306 (IF: 8.9), 10.1109/TKDE.2022.3222538, 2023
- **Tomas Petricek**. The Gamma: Programmatic Data Exploration for Non-programmers. VL/HCC '22: Proceedings of the IEEE Symposium on Visual Languages and Human-Centric Computing, 10.1109/VL/HCC53370.2022.9833134, 2022
- Roly Perera, Minh Nguyen, **Tomas Petricek** and Meng Wang. Linked Visualisations via Galois Dependencies. Proceedings of the ACM on Programming Languages, vol. 6 (POPL), pp. 1-29 (IF: 1.8), 10.1145/3498668, 2022
- **Tomas Petricek**. Composable Data Visualizations. Journal of Functional Programming, vol. 31, e. 13, Cambridge University Press (IF: 1.1), 10.1017/S0956796821000046, 2021